Exercise Sheet 2

Probability, Data and a Parameter

Probabilistic Machine Learning

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2.A Conditioning and Bayes

Exercise 2.A.1: Consequences of the Axioms

Using only the three Kolmogorov axioms (and set algebra), show:

1. $P(A \setminus B) = P(A) - P(A \cap B)$
2. $P(A \cup B) \leq P(A) + P(B)$
3. If $A \subseteq B$, then $P(A) \leq P(B)$.

Exercise 2.A.1: Consequences of the Axioms — Solution

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1. $A = (A \setminus B) \cup (A \cap B)$, disjoint, so by Axiom III $P(A) = P(A \setminus B) + P(A \cap B)$.
2. $A \cup B = A \cup (B \setminus A)$, disjoint: $P(A \cup B) = P(A) + P(B \setminus A) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B)$, since $P(A \cap B) \geq 0$ (Axiom I).
3. $B = A \cup (B \setminus A)$, disjoint: $P(B) = P(A) + P(B \setminus A) \geq P(A)$ by Axioms III and I.

Exercise 2.A.2: One Flag, Two Flags

Fraud occurs in **0.2%** of transactions. A rule flags **95%** of frauds and **2%** of legitimate transactions.

1. What fraction of flagged transactions is fraud?
2. A second, different rule has the same rates. Assume the two rules err **independently given the class**. A transaction is flagged by both. Now?
3. When would the assumption in 2 be badly wrong — and in which direction would the answer then be off?

Exercise 2.A.2: One Flag, Two Flags — Solution

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1. $P(\text{flag}) = 0.002 \cdot 0.95 + 0.998 \cdot 0.02 = 0.02186$, so $P(\text{fraud} \mid \text{flag}) = 0.0019 / 0.02186 \approx 8.7\%$.
2. In odds: prior odds $0.002 / 0.998 \approx 0.0020$; each flag multiplies them by the likelihood ratio $0.95 / 0.02 = 47.5$. Two flags: $0.0020 \cdot 47.5^2 \approx 4.52$, i.e. $P \approx 4.52 / 5.52 \approx 82\%$.
3. If both rules look at the same signal (say, both react to foreign IPs), the second flag adds little new evidence, yet the formula counts it twice — the answer is **too confident**. This is exactly the assumption naive Bayes makes.

Exercise 2.A.3: Independent Is Not Disjoint

1. Show that two disjoint events A, B with $P(A), P(B) > 0$ are never independent.
2. Flip two fair coins. Let A = "first is heads", B = "second is heads", C = "the coins agree". Show that A, B, C are pairwise independent but not mutually independent.

Exercise 2.A.3: Independent Is Not Disjoint — Solution

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1. $P(A \cap B) = 0 \neq P(A)\,P(B) > 0$.
2. Each event has probability $\frac{1}{2}$. $A \cap B = \{HH\}$, $A \cap C = \{HH\}$, $B \cap C = \{HH\}$: each has probability $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$ — pairwise independent. But $A \cap B \cap C = \{HH\}$ has probability $\frac{1}{4} \neq \frac{1}{8}$. Knowing A and B determines C.

2.B Random Variables and the Joint

Exercise 2.B.1: App Use and Churn

The joint distribution of X (uses the app: 1/0) and Y (churns: 1/0):

	$Y = 0$	$Y = 1$
$X = 0$	0.30	0.20
$X = 1$	0.40	0.10

1. Compute $P(X = 1)$, $P(Y = 1)$, $P(Y = 1 \mid X = 1)$ and $P(Y = 1 \mid X = 0)$.
2. Are X and Y independent? Compute $\text{cov}(X, Y)$.
3. Does the table show that the app *reduces* churn? What else could produce it?

Exercise 2.B.1: App Use and Churn — Solution

Exercise 2.B.1: App Use and Churn — Solution

1. $P(X{=}1) = 0.5$, $P(Y{=}1) = 0.3$, $P(Y{=}1 \mid X{=}1) = 0.1/0.5 = 0.2$, $P(Y{=}1 \mid X{=}0) = 0.2/0.5 = 0.4$.
2. Not independent ($0.2 \neq 0.3$). $\mathrm{cov}(X, Y) = E[XY] - E[X]E[Y] = 0.10 - 0.5 \cdot 0.3 = -0.05$.
3. No. A joint distribution shows association, not direction: loyal customers may be the ones who install the app (the abduction exercise 1.C.1 of Exercise Sheet 1). Deciding needs an experiment or a model of the causes.

Exercise 2.B.2: Dependent, Yet Uncorrelated

Let X take the values $-1, 0, 1$ with probability $\frac{1}{3}$ each, and $Y = X^2$.

1. Compute $E[X]$, $E[Y]$ and $\mathrm{cov}(X, Y)$.
2. Show that X and Y are not independent.
3. What does this say about using correlation to screen features?

Exercise 2.B.2: Dependent, Yet Uncorrelated — Solution

Exercise 2.B.2: Dependent, Yet Uncorrelated — Solution

1. $E[X] = 0, E[Y] = \frac{2}{3}, E[XY] = E[X^3] = 0$, so $\operatorname{cov} = 0$.
2. $P(Y = 0 \mid X = 0) = 1 \neq P(Y = 0) = \frac{1}{3}$.
3. A feature can determine the target completely and still have zero correlation with it. Correlation screens only for (monotone) linear association.

2.C Data, Statistics and a Parameter

Exercise 2.C.1: Scale and Model

For each attribute of a customer table, give the scale and a first probabilistic model, and name one statistic that would be **invalid**: payment method (card / invoice / PayPal) · delivery rating (1–5 stars) · number of orders last year · order value in € · has a newsletter subscription.

Exercise 2.C.1: Scale and Model — Solution

Exercise 2.C.1: Scale and Model — Solution

attribute	scale	first model	invalid
payment method	nominal	categorical (p_1, p_2, p_3)	mean, median
rating	ordinal	categorical (order unused) or ordinal model	mean (gaps unknown) — defensible only with care
orders	ratio (count)	Poisson, or binomial with a maximum	— (all valid)
order value	ratio, positive, skewed	log-normal	none, but the mean is outlier-sensitive
newsletter	nominal, binary	Bernoulli(p)	— (coded 0/1, the mean *is* $\hat{p}$)

Exercise 2.C.2: How Precise Is a Conversion Rate?

A campaign reached **400** users; **60** converted.

1. Estimate the conversion rate and its standard error.
2. Give an approximate 95% confidence interval and state correctly what the 95% refers to.
3. How many users would give a half-width of ± 0.01?

Exercise 2.C.2: How Precise Is a Conversion Rate? — Solution

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1. $\hat{p} = 60/400 = 0.15; \mathrm{SE} = \sqrt{0.15 \cdot 0.85/400} \approx 0.018$.
2. $0.15 \pm 1.96 \cdot 0.018 \approx [0.115, 0.185]$. 95% of intervals built **by this procedure** contain the true rate — not "95% probability that the rate lies in this interval".
3. $1.96\sqrt{0.1275/n} = 0.01 \Rightarrow n \approx 4{,}900$ — twelve times the data for a 3.5 times narrower interval: precision costs $\sqrt{n}$.

Exercise 2.C.3: Two Analysts, One Campaign

Same data: 60 of 400 converted. Analyst A uses a flat prior $\mathrm{Beta}(1, 1)$; analyst B, from earlier campaigns, uses $\mathrm{Beta}(20, 80)$.

1. Write down both posteriors and their means.
2. How many "pseudo-observations" is B's prior worth? Why do the two analysts nearly agree?
3. The data arrived in two batches: 25 of the first 200, 35 of the second 200. Show that updating twice gives the same posterior as updating once.
4. What is A's probability that the next user converts?

Exercise 2.C.3: Two Analysts, One Campaign — Solution

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1. A: $\mathrm{Beta}(61, 341)$, mean $61/402 \approx 0.152$. B: $\mathrm{Beta}(80, 420)$, mean $80/500 = 0.160$. 95% credible intervals: A $[0.118, 0.188]$, B $[0.129, 0.193]$.
2. $a + b = 100$ pseudo-observations against 400 real ones; B's prior mean 0.2 is pulled almost all the way to the data's 0.15.
3. A: $\mathrm{Beta}(1, 1) \to \mathrm{Beta}(26, 176) \to \mathrm{Beta}(61, 341)$ — counts simply add, in any order.
4. The posterior mean, ≈ 0.152: averaging θ over the posterior.

Exercise 2.C.4: A Sampling Distribution by Simulation (Python)

```python
import numpy as np
rng = np.random.default_rng(0)
samples = rng.exponential(scale=2.0, size=(1000, 25))   # 1000 samples, n = 25
```

1. Compute the mean of each sample and the standard deviation of those means. Compare with $\sigma/\sqrt{n}$ (for this distribution $\sigma = 2$).
2. The exponential is strongly skewed. Plot a histogram of the means — what shape do you see, and which theorem predicts it?
3. Repeat with $n = 100$. By what factor does the spread shrink?

Exercise 2.C.4: A Sampling Distribution by Simulation — Solution

Exercise 2.C.4: A Sampling Distribution by Simulation — Solution

```
means = samples.mean(axis=1)
print(means.std(ddof=1), 2.0 / np.sqrt(25))   # ≈ 0.40 vs 0.40
```

1. About 0.40 against $\sigma/\sqrt{n} = 2/5 = 0.40$.
2. Close to a bell, slightly right-skewed — the central limit theorem, not yet perfect at $n = 25$ for a skewed population.
3. Four times the data, half the spread: about 0.20.